

Exam 1 Review

Dr. Wang

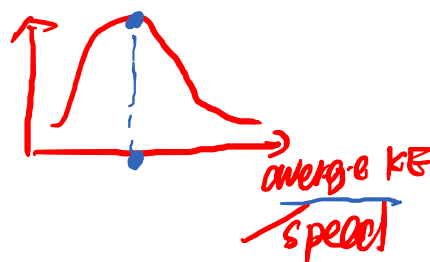
Energy, heat and work (qualitative analysis)

- Determine the **system** and surroundings for chemical reactions.

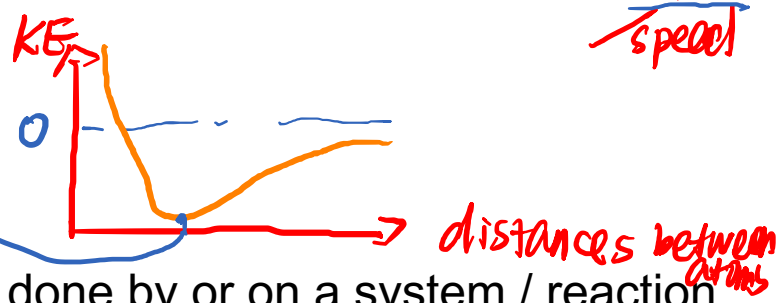
- Relate KE, PE, and internal energy for open, closed, and isolated systems.

{ Kinetic energy: motion $\Rightarrow$ thermal energy #
 potential energy: (position / composition) $\Rightarrow$ chemical PE

- Relate temperature and average molecular KE.



chemical bond formation



- Relate the volume change of the gas to the work done by or on a system / reaction.

(constant pressure)

work: $w = -P_{\text{ext}} \Delta V$

$\Delta V \uparrow$ ($\Delta n_{\text{gas}} \uparrow$) , $w (-)$, work done by system
 $\Delta V \downarrow$ ($\Delta n_{\text{gas}} \downarrow$) , $w (+)$, work done on system

Your turn!

1. Which of the following signs on q and w represent a system that is doing work on the surroundings, as well as losing heat to the surroundings?

☒ A) $q = -$, $w = -$

B) $q = +$, $w = +$

C) $q = -$, $w = +$

D) $q = +$, $w = -$

2. Which of the following processes is endothermic? The system is underlined.

A) the freezing of water (-)

B) the combustion of diesel (-)

C) a hot cup of soup cools on a countertop (-)

D) the chemical reaction in a "hot pack" often used to treat sore muscles (-)

☒ E) the vaporization of ethanol (+)



Heat and Calorimeter (quantitative analysis)

- Quantify heat transfer from one object to another using $q = mC_s\Delta T$ or $q = C\Delta T$.

$$C_s: \frac{J}{g \cdot ^\circ C}$$

$$C: \frac{J}{^\circ C}$$

- Quantify the heat for a chemical reaction using data from a ~~bomb~~ or coffee cup calorimeter. (constant pressure) w $\begin{cases} \Delta n_{gas} > 0, w(-) \\ \Delta n_{gas} = 0, w=0 \\ \Delta n_{gas} < 0, w(+)\end{cases}$

$$q_{rxn} = -q_{cal} = -m(\text{everything in calorimeter}) C_s \Delta T$$

$$\Delta H_{rxn} = \frac{q_{rxn}}{\#mol}$$

- Relate the ΔE_{rxn} with heat and work for a chemical reaction.

$$\Delta E_{rxn} = q_{rxn} + w$$

- Relate the amount of heat released or absorbed by a reaction using a balanced chemical reaction and the reaction stoichiometry.
- Use stoichiometry to relate mass and mole amounts for a balanced chemical reaction

Your turn!

Substances A and B, initially at different temperatures, come in contact with each other and reach **thermal equilibrium**. The mass of substance B is twice the mass of substance A. The **specific heat capacity** of substance B is **twice** the **specific heat capacity** of substance A. Which statement is true about the final temperature of the two substances once thermal equilibrium is reached?

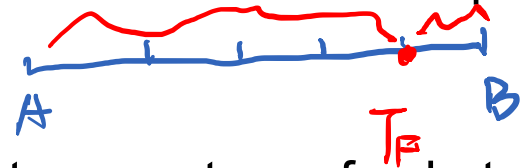
$$\cancel{m} \cdot \cancel{C_s} \cdot \Delta T = \cancel{2m} \cdot \cancel{2C_s} \cdot \Delta T$$

q_A q_B

A) The final temperature is closer to the initial temperature of substance B than it is to the initial temperature of substance A.

$$\Delta T(A) = 4 \Delta T(B)$$

B) The final temperature is exactly midway between the initial temperatures of substances A and B.



C) The final temperature is closer to the initial temperature of substance A than it is to the initial temperature of substance B

thermal equilibrium

$$\begin{cases} q_A + q_B = 0 & ; & q_A = -q_B \\ T_F(A) = T_F(B) \end{cases}$$



Your turn!

A 4.82 g lead weight, initially at 11.2°C , is submerged in 8.46 g of water at 54.6°C in an insulated container. What is the final temperature of both substances at thermal equilibrium? (The specific heat capacity of lead is $0.128 \text{ J/g} \cdot ^{\circ}\text{C}$ and that of water is $4.18 \text{ J/g} \cdot ^{\circ}\text{C}$.)

Your turn!

A 4.82 g lead weight, initially at 11.2°C, is submerged in 8.46 g of water at 54.6°C in an insulated container. What is the final temperature of both substances at thermal equilibrium? (The specific heat capacity of lead is 0.128 J/g • °C and that of water is 4.18 J/g • °C.)

$$q_{\text{Pb}} + q_{\text{H}_2\text{O}} = 0$$

$$4.82\text{g} \times \frac{0.128\text{J}}{\text{g} \cdot ^\circ\text{C}} \times (T_F - 11.2^\circ\text{C}) + 8.46\text{g} \times \frac{4.18\text{J}}{\text{g} \cdot ^\circ\text{C}} \times (T_F - 54.6^\circ\text{C}) = 0$$

$$\frac{0.61696\text{J}}{^\circ\text{C}} \times (T_F - 11.2^\circ\text{C}) + \frac{35.3628\text{J}}{^\circ\text{C}} \times (T_F - 54.6^\circ\text{C}) = 0$$

$$\frac{35.97896\text{J}}{^\circ\text{C}} T_F = 1937.675152\text{J}$$

$$T_F = 53.9^\circ\text{C}$$

Announcement 2/25

Exam 1 at 11 AM on 2/27:

- Due Date:
(2nd bonus point: midnight 2/25; 3rd +4th bonus point: 10 AM 2/27)
- 50 minutes long
- Calculators (any kind), pens/pencils/highlighters, and erasers are allowed.
- Learning objective uploaded

Chem RePS Session

Wed 2:00 - 3:30 PM at KC 230

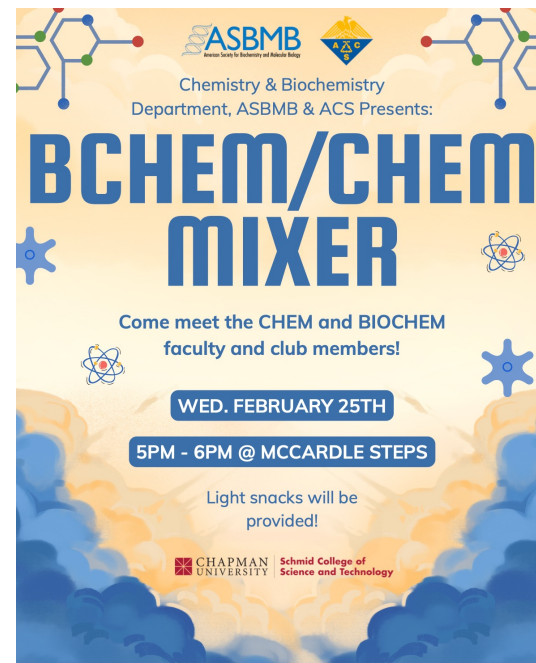
Office Hour

- Dr. Wang's: Thu 10-11 AM on Zoom

SI Session

Saige Douglass – Wed 5:30 – 7:30 PM at Hashinger 412

Paige Kwan - Thu 5 - 7 PM at KC134



Your turn!

$$q > 0 (+)$$

Two aqueous solutions are both at room temperature and are then mixed in a coffee cup calorimeter. The reaction causes the temperature of the resulting solution to fall below room temperature. Which of the following statements is TRUE?



Surr. calorimeter (-)

Sys. rxn (+)

~~A)~~ The products have a lower potential energy than the reactants.

~~B)~~ The heat change of the experiment is the energy change of the reaction. $w = 0$

~~C)~~ The reaction is exothermic.

$$\Delta E = q + w$$

$(-)$ $(+)$ $(-)$

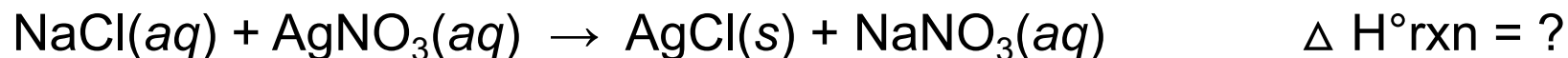
~~D)~~ Energy is leaving the system during reaction.

E) None of the above statements is true.



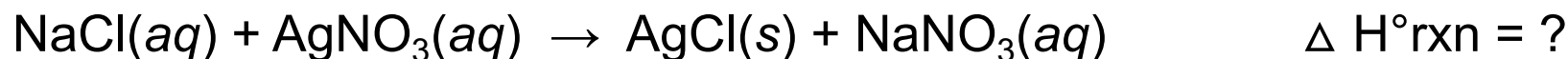
Your turn!

Two solutions, initially at 24.60°C, are mixed in a coffee cup calorimeter. When a 100.0 mL volume of 0.100 M AgNO₃ solution is mixed with a 100.0 mL sample of 0.200 M NaCl solution, the temperature in the calorimeter rises to 25.30°C. Determine the $\Delta H^{\circ}\text{rxn}$ for the reaction as written below. Assume that the density and heat capacity of the solutions is the same as that of water (1.00g/mL). The specific heat capacity of water = 4.18 J/g°C.



Your turn!

Two solutions, initially at 24.60°C , are mixed in a coffee cup calorimeter. When a 100.0 mL volume of 0.100 M AgNO_3 solution is mixed with a 100.0 mL sample of 0.200 M NaCl solution, the temperature in the calorimeter rises to 25.30°C . Determine the $\Delta H^{\circ}\text{rxn}$ for the reaction as written below. Assume that the density and heat capacity of the solutions is the same as that of water (1.00 g/mL). The specific heat capacity of water = $4.18\text{ J/g}^{\circ}\text{C}$.



$$q_{\text{rxn}} = -q_{\text{calorimeter}} = - (100.0\text{ mL} + 100.0\text{ mL}) \times \frac{1.00\text{ g}}{\text{mL}} \times \frac{4.18\text{ J}}{\text{g}^{\circ}\text{C}} \times (25.30^{\circ}\text{C} - 24.60^{\circ}\text{C})$$

$$= -590\text{ J}$$

$$n(\text{AgNO}_3) = 100.0\text{ mL} \times \frac{0.100\text{ mol AgNO}_3}{1\text{ L}} \times \frac{1\text{ L}}{10^{-3}\text{ L}} = 0.01\text{ mol}$$

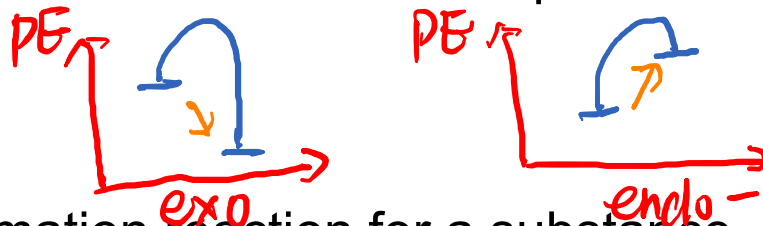
$$n(\text{NaCl}) = 100.0\text{ mL} \times \frac{0.200\text{ mol NaCl}}{1\text{ L}} \times \frac{1\text{ L}}{10^{-3}\text{ L}} = 0.02\text{ mol}$$

If AgNO_3 is all used up. it needs $0.01\text{ mol AgNO}_3 \times \frac{1\text{ NaCl}}{1\text{ AgNO}_3} = 0.01\text{ mol NaCl} < 0.02\text{ mol}$
 AgNO_3 is LR

$$\Delta H_{\text{rxn}} = \frac{-590\text{ J}}{0.01\text{ mol}} \times \frac{1\text{ kJ}}{10^3\text{ J}} = -59\text{ kJ}$$

Enthalpy and Hess's Law

- Relate the chemical PE of the reactants and products for an exothermic or endothermic reaction.



- Write the standard formation reaction for a substance.

elements ($\Delta H_f^\circ = 0 \frac{\text{kJ}}{\text{mol}}$) $\rightarrow$ 1 substance $\Delta H_{\text{rxn}}^\circ = \Delta H_f^\circ(\text{substance})$

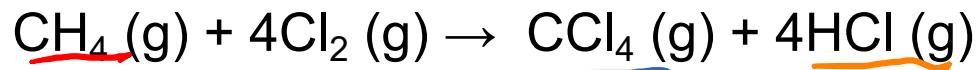
- Determine the ΔH_{rxn} for a reaction from ΔH_f° data table.

$$\Delta H_{\text{rxn}} = \Delta H_f^\circ(\text{product}) - \Delta H_f^\circ(\text{reactant})$$

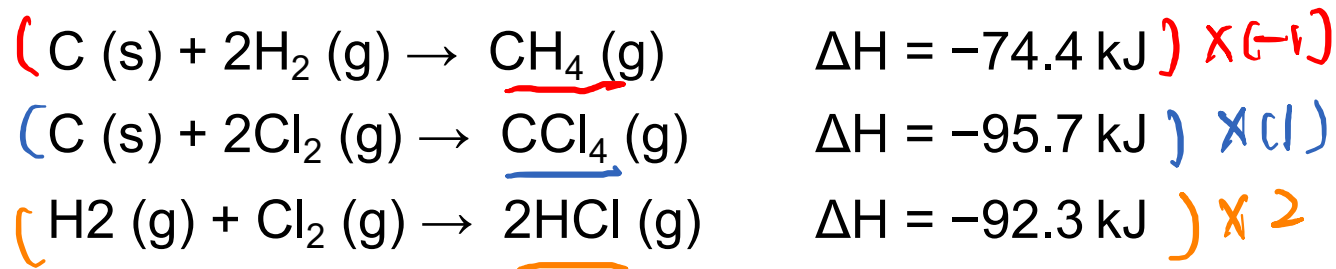
- Determine the ΔH_{rxn} from related chemical equations using Hess's Law.

Your turn!

Calculate $\Delta H^\circ_{\text{rxn}}$ for the following reaction:

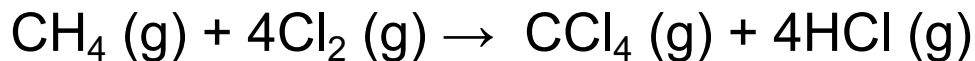


Use the following reactions and given ΔH 's:

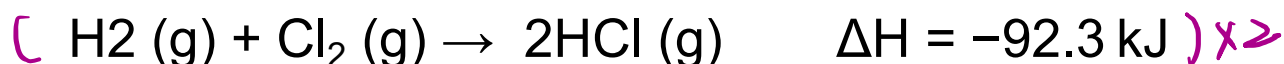
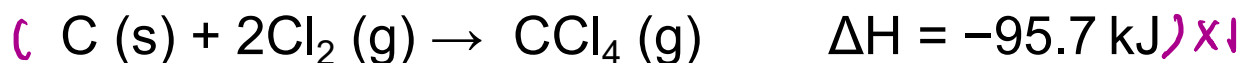
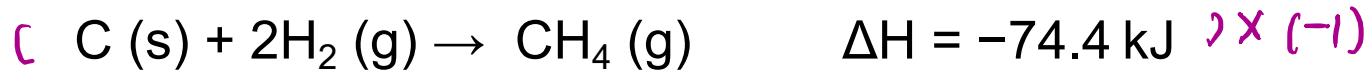


Your turn!

Calculate $\Delta H^\circ_{\text{rxn}}$ for the following reaction:



Use the following reactions and given ΔH 's:



$$\Delta H = (-74.4 \text{ kJ}) \times (-1) + (-95.7 \text{ kJ}) \times 1 + (-92.3 \text{ kJ}) \times 2 = -205.9 \text{ kJ}$$

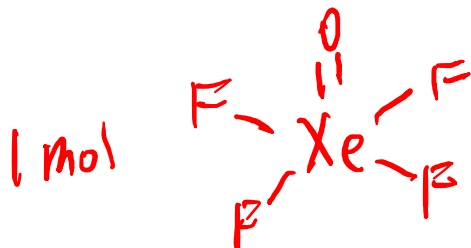
work

Enthalpy and Bond Energy

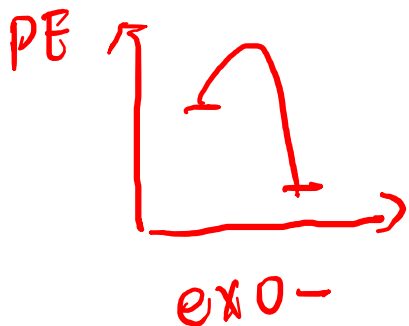
- Use bond energies for a chemical reaction to calculate an approximate ΔH for a chemical reaction.

$$\Delta H_{rxn} = (+)BE \text{ in reactants} + (-) BE \text{ in products}$$

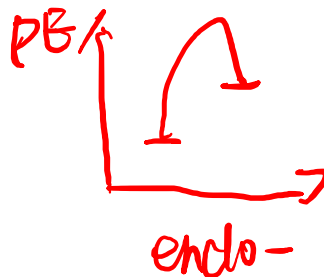
- Use Lewis structures to determine ΔH using bond energies.



- Relate the bond strength of the reactants and products for an exothermic or endothermic reaction. $\rightarrow$ stronger = stable = low PE



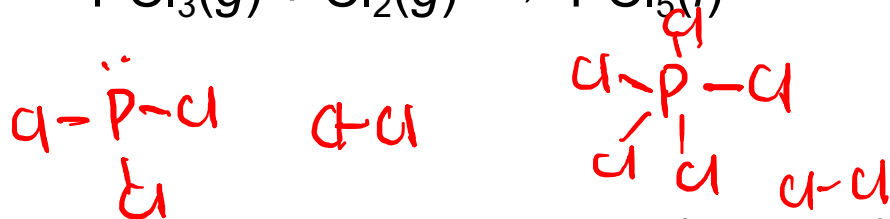
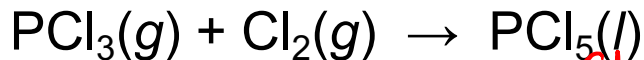
weaker $\rightarrow$ stronger



stronger $\rightarrow$ weaker

Your turn!

Use the bond energies provided to estimate $\Delta H^\circ_{\text{rxn}}$ for the reaction below. Which coefficients are correct?



Bond	Bond Energy (kJ/mol)
Cl-Cl	243
P-Cl	331

$$\Delta H^\circ_{\text{rxn}} = n \text{ mol} * \underline{243 \text{ kJ/mol}} + m \text{ mol} * 331 \text{ kJ/mol}$$

A) $n = -1; m = 2$

C) $n = -2; m = 2$

E) $n = 1; m = -2$

G) $n = 2; m = -2$

B) $n = -1; m = 8$

D) $n = -2; m = 8$

F) $n = 1; m = -8$

H) $n = 2; m = -8$

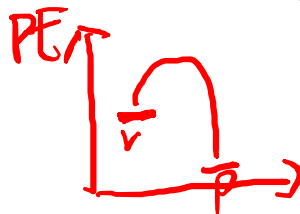


This reaction is exothermic reaction. The chemical potential energy of reactant is _____ the products.

A) Larger than

B) less than

C) equals to



Entropy and Gibbs Free Energy

- Relate spontaneity to the ΔG , ΔH , and ΔS for a reaction.
- Describe the contribution of ΔS and ΔH on the spontaneity of a reaction.

$$\Delta G = \Delta H - T\Delta S$$

ΔH	ΔS	ΔG
(-)	(+)	(-)
(+)	(-)	(+)
$\Rightarrow$ (+)	(+)	high T (-)
$\Rightarrow$ (-)	(-)	low T (-)

0
 $\neq$

} 25°C 298K
 1 atm
 1 M

- Predict the sign of ΔS for a chemical reaction.

$$\Delta S \uparrow \left\{ \begin{array}{l} \Delta V \uparrow \\ \Delta n \uparrow \\ \Delta T \uparrow \\ \text{gas} \gg \text{aq} > \text{l} > \text{(s)} \\ \text{complex structure} \uparrow \end{array} \right.$$

- Quantify ΔG at a specific temperature for a reaction with the known ΔH and ΔS .
- Use the ΔG for a chemical reaction to describe the amount of usable (free) energy available to the surroundings to do work.

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G_{\text{rxn}}^{\ominus} = \Delta G_f^{\ominus} - \Delta G_f^{\ominus}$$

ΔG

(-) spontaneous
 = 0 equilibrium
 (+) nonspontaneous

Your turn!

1. Identify the process in which the entropy decreases.

A) an increase in the number of moles of a gas during a chemical reaction

☒ B) the phase transition from a gas to a liquid

C) the phase transition from a solid to a gas

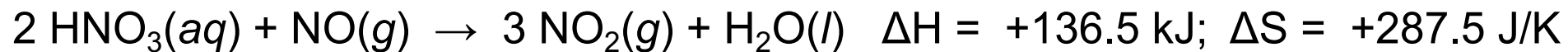
D) the phase transition from a solid to a liquid

E) the phase transition from a liquid to a gas



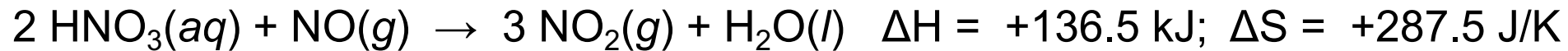
Your turn! $\Delta G = \Delta H - T\Delta S$

Below what temperature does the following reaction become nonspontaneous?



Your turn! $\Delta G = \Delta H - T\Delta S$

Below what temperature does the following reaction become nonspontaneous?



$$+136.5 \text{ kJ} - T \times \left(+287.5 \frac{\text{J}}{\text{K}} \right) \times \frac{1 \text{ kJ}}{10^3 \text{ J}} = 0$$

$$T = 475 \text{ K}$$

Your turn! $\Delta G = \Delta H - T\Delta S$

Calculate the $\Delta G^\circ_{\text{rxn}}$ using the following information.



ΔH°_f (kJ/mol)	-133.9	50.6	0	-285.8
S° (J/mol·K)	266.9	121.2	191.6	70.0

Your turn! $\Delta G = \Delta H - T\Delta S$

Calculate the $\Delta G^\circ_{\text{rxn}}$ using the following information.



ΔH°_f (kJ/mol)	-133.9	50.6	0	-285.8
-----------------------------	--------	------	---	--------

S° (J/mol·K)	266.9	121.2	191.6	70.0
---------------------	-------	-------	-------	------

$$\Delta H = 12 \text{ mol H}_2\text{O} \times \frac{-285.8 \text{ kJ}}{1 \text{ mol H}_2\text{O}} + 7 \text{ mol N}_2 \times \frac{0 \text{ kJ}}{1 \text{ mol N}_2} - \left(5 \text{ mol N}_2\text{H}_4 \times \frac{50.6 \text{ kJ}}{1 \text{ mol N}_2\text{H}_4} + 4 \text{ mol HNO}_3 \times \frac{-133.9 \text{ kJ}}{1 \text{ mol HNO}_3} \right)$$

$$= -3147 \text{ kJ}$$

$$\Delta S = 12 \text{ mol H}_2\text{O} \times \frac{70.0 \text{ J}}{1 \text{ mol H}_2\text{O} \cdot \text{K}} + 7 \text{ mol N}_2 \times \frac{191.6 \text{ J}}{1 \text{ mol N}_2 \cdot \text{K}} - \left(5 \text{ mol N}_2\text{H}_4 \times \frac{121.2 \text{ J}}{1 \text{ mol N}_2\text{H}_4 \cdot \text{K}} + 4 \text{ mol HNO}_3 \times \frac{266.9 \text{ J}}{1 \text{ mol HNO}_3 \cdot \text{K}} \right)$$

$$= 890.8 \frac{\text{J}}{\text{K}}$$

$$\Delta G^\circ_{\text{rxn}} = -3147 \text{ kJ} - (25 + 273.15) \text{ K} \cdot \frac{890.8 \text{ J}}{\text{K}} \times \frac{1 \text{ kJ}}{10^3 \text{ J}} = -3.70 \times 10^3 \text{ kJ}$$